

JIEWAN HONG

Baltimore, MD | jhong68@jhu.edu | github.com/jiewan02 | jiewan02.github.io/portfolio | [LinkedIn](#)

EDUCATION

Johns Hopkins University

Baltimore, MD | Expected May 2027

B.S. in Computer Science; second major in Cognitive Science

- **Coursework:** Machine Learning; Deep Learning; Intro Algorithms; Data Structures; Intermediate Programming; Linear Algebra; Human-Computer Interaction; Computational Cognitive Neuroscience of Vision
- **Fall 2026:** Computer System Fundamentals; Artificial Intelligence; Visual Cognition, Cognitive Neuroimaging
- 2023–2025 leave for mandatory Republic of Korea military service; completed AI and full-stack training (below)

TECHNICAL SKILLS

- **Languages:** Python, Java, SQL, basic familiarity with C/C++/TypeScript
- **AI / ML:** PyTorch, YOLO, LLM integration & prompt engineering, retrieval-augmented generation (graph RAG, Neo4j)
- **Web & Tools:** React, NestJS, Node.js, REST APIs, Git/GitHub

RESEARCH EXPERIENCE

Undergraduate Research Assistant — GLINT Lab, Johns Hopkins University

Baltimore, MD | Sep 2026 – Present

- Building a benchmark of LLM color knowledge with Profs. Jennifer Hu and Marina Bedny, adapted from blind and sighted adult studies

Undergraduate Researcher — Intersection Safety Research, Johns Hopkins University

Baltimore, MD | May 2026 – Present

- Analyzed roadside LiDAR data covering millions of vehicle movements at signalized intersections to surface recurring near-collision patterns before serious accidents occur — for the Prince George's County Dept. of Public Works & Transportation
- Characterized vehicle–pedestrian conflict events to trace the conditions that repeatedly generate near-misses
- Presented findings and signal-timing recommendations to county staff, backed by an automated pipeline that re-runs as data arrives
- Leading a Fall 2026 independent study applying FHWA/AASHTO sight-distance criteria to link limited visibility to near-miss clusters

Research Assistant — Isik Lab, Johns Hopkins University

Baltimore, MD | Jan 2023 – Jun 2023

- Preprocessed and analyzed EEG and intracranial neural recordings for studies of human cognition and social perception

EXPERIENCE

Software Engineering Intern — [Manyfast](#)

Seoul, South Korea | Jun 2026 – Aug 2026

- Built features for an AI-powered web platform where teams design and edit app wireframes together in real time, shipping 30+ improvements across the user-facing product, admin tools, and server (TypeScript, React, NestJS)
- Redesigned shared project links so outside viewers see the same live canvas as editors, with updates and comments appearing instantly, while removing substantial duplicated code
- Built the platform's overload protection — automatic request throttling, per-team usage limits, and continued operation when parts of the infrastructure fail; hardened through nine rounds of code review

AI Project Developer Trainee — Alpaco Campus (AI intensive program)

Seoul, South Korea | Aug 2025 – Dec 2025

- Completed an intensive, project-based AI engineering program; designed and built two AI products end to end (described below)

Full-Stack Developer Trainee — Korea Software Technology Association

Seoul, South Korea | Feb 2025 – Jul 2025

- Delivered features across the frontend, backend, and database of three team-built web applications — REST APIs with authentication, content creation, and search — along with relational data modeling and multi-user business logic

Military Researcher, Sergeant — ROK Army Logistics Command

Daejeon, South Korea | Jun 2023 – Dec 2024

- Contributed to AI and data-analysis tools that improved army logistics resource allocation and planning
- Built text-analysis workflows that categorized and summarized civilian complaints for leadership

PROJECTS

- **EduScope — classroom engagement analytics (computer vision, YOLO):** detects phone use, gaze direction, and hand-raising in live classroom video, producing per-session engagement analytics for instructors — [portfolio](#)
- **Eat Smart! — recipe recommendation (LLM + Neo4j graph RAG):** grounds LLM recipe recommendations in a knowledge graph; integrated a React frontend, Node.js/Express backend, and Flask model server — [slides](#)

HONORS, ACTIVITIES & LANGUAGES

- **Honors:** Dean's List, JHU
- **Activities:** Tutor, JHU First Generation Investors; Marketing Team, JHU KROSS (Korean-Christian ministry); JHU Wind Ensemble
- **Languages:** English (native), Korean (native), Mandarin Chinese (intermediate, HSK Level 5)